THE PURELY INSEPARABLE ELEMENTS OF AN EXTENSION FORM A SUBEXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The purely inseparable analogue of lemma-algebraic-elements (b7 companion card L48374); together the two subextensions frame the separable/purely-inseparable decomposition of the chapter.

Editorial note. The label lemma-purely-inseparable-elements names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Freshman's dream in characteristic p: for a p-power q, (alpha + beta)^q = alpha^q + beta^q, since the intermediate binomial coefficients are divisible by p
- Any two p-powers q, q' are bounded by a common p-power q" (for instance q q')

Lemma 0.1. *Let E/k be a field extension. Then the elements of E purely-inseparable over k form a subextension of E/k.*

Proof. Let p be the characteristic of k. Let $\alpha, \beta \in E$ be purely inseparable over k. Say $\alpha^q \in k$ and $\beta^{q'} \in k$ for some p-powers q, q'. If q'' is a p-power, then $(\alpha + \beta)^{q''} = \alpha^{q''} + \beta^{q''}$. Hence if $q'' \geq q, q'$, then we conclude that $\alpha + \beta$ is purely inseparable over k. Similarly for the difference, product and quotient of α and β. $\square$

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